

Rung 6 — What can even *be* certified? (the summit)

Where you start: you've bounded a fixed-point sine, composed certificates, and proven a control loop safe forever. Every one of those rested on the functions involved being *well-behaved*. This rung asks the question underneath the whole project: **which functions are well-behaved enough to certify at all — and is there a boundary where "provable" runs out?**

Honesty up front: this is real research mathematics — model theory and real algebraic geometry. This lesson is a **map of the peak, not the climb**. The goal is to see the shape of the ideas and why they matter to us, then point you at the trail (which is long). Nobody summits this in a weekend, and that's fine — it's the frontier the research side of Monogate actually works on.

1. The bench

Two facts from earlier rungs, now viewed from high up:

- `sin`'s Taylor remainder was boundable (Rung 2) because `sin` and all its derivatives stay small and orderly.
- The control loop was boundable (Rung 5) because the plant was a *contraction* — orderly decay.

But not every function is orderly. Some wiggle infinitely, or have infinitely many zeros in a finite window — for those, no finite certificate exists. So there's a real line between "tame" (certifiable) and "wild" (not). This rung is about where that line is.

2. The one new idea: *tameness* — finiteness you can count on

The organizing concept is **o-minimality** (order-minimality). A collection of functions/shapes is *o-minimal* if every set you can describe with it, on a line, is just **finitely many points and intervals** — no infinitely-wiggling, no pathological dust. That finiteness is exactly what a certificate needs: finitely many cases, a worst case that exists, a bound you can name.

The remarkable facts (each a deep theorem):

- **Polynomials** are tame.
- `exp` added to polynomials is *still* tame (Wilkie's theorem — a landmark result).
- `sin` **on a bounded interval** is tame (which is why our `[0,1]` domain matters — `sin` over the *whole* line is wild, with infinitely many zeros; restricted, it's tame). This is the deep reason the certificate had a domain restriction, and why the certified NCO needed range reduction.

So "tame" isn't hand-waving — it's a precise property, and a huge chunk of 20th-century logic went into proving which everyday functions have it.

3. Do it by hand — well, follow the idea by hand

You won't derive o-minimality by hand, but you *can* hold the key counting idea, **Khovanskii's fewnomial theorem**:

The number of solutions of a system of "simple" transcendental equations is bounded by their **complexity** — how many terms and how they're built — **not by their degree**.

Contrast with intuition from polynomials, where more degree can mean more roots. Khovanskii says: for the right tame functions, a *simple* equation has *few* solutions no matter how the numbers inside blow up. That "few, and you can bound how few" is the engine that makes finiteness — hence certifiability — go through. **Pfaffian functions** (ones that satisfy a triangular system of differential equations — `exp`, `sin / cos` on bounded domains, and chains built from them) are the concrete class Khovanskii's bounds apply to. The research repo's "exponential-tower" work lives right here: bounding how many zeros a tower `exp(exp(...))` can have.

4. Make it formal

This is the live edge. Formalizing o-minimality and fewnomial bounds in Lean is hard and partial — it's exactly the kind of thing the research side pushes on (e.g. proving a depth-`N` exponential tower has finitely many zeros, `sorryAx`-free). The point for you: the *same* discipline you learned on Rung 4 — no gaps, no borrowed axioms — is being applied at the frontier of what's been formalized at all. You now understand, at least in outline, what those proofs are *for*.

5. The view from the top

Look back down the ladder you climbed:

You learned	so you could answer
fixed-point error (R1)	how wrong is one multiply
reading proofs (R1.5)	what "for all x" even means
Taylor's remainder (R2)	how wrong the approximation is
composing bounds (R3)	how errors stack in a system
Lean (R4)	how a machine checks there are no gaps
Lyapunov (R5)	why a loop is safe forever
o-minimality (R6)	which functions can be certified at all

That's the whole arc — **from "how wrong is one hardware multiply" all the way up to "what can even be known."** It's the same question asked at deeper and deeper levels, and the answer, at every level, was made

honest by a proof and confirmed on a real bench.

Where to actually climb

- Surveys on **Khovanskii's theorem / fewnomials** for the ideas before the machinery.
- **A. Khovanskii, *Fewnomials*** — the source.
- **L. van den Dries, *Tame Topology and O-minimal Structures*** — the classic on o-minimality (hard; a long-term companion, not a first read).
- The Monogate research repo's exponential-tower / operator-basis work — where this stops being a textbook and becomes new proofs.

You've reached the top of the built ladder. From here, the trail is open — and it's the same one the research agents are cutting.

Exercises & checkpoint

1. Explain in one or two sentences why `sin` on `[0,1]` is "tame" (certifiable) but `sin` on *all* of the real line is not. (Hint: count the zeros.)
2. One sentence: Khovanskii's theorem bounds the number of solutions of a simple transcendental system by *what* — and not by what?
3. (Stretch, no pressure) Skim a fewnomials survey. Write down one sentence you understood and one you didn't. The second one is your next trailhead.

Checkpoint: you can explain why a domain restriction (like `[0,1]`) keeps showing up in our certificates — it's tameness, made concrete.